

Question 1 (1 point)

The emptiness problem for Turing machines — deciding whether $L(M) = \emptyset$ — is undecidable and not recognizable because we cannot confirm in finite time that a TM accepts no strings. For NFAs and CFGs, emptiness is decidable, so options B and C are recognizable. Option A is undecidable but recognizable in some cases, whereas option D is explicitly not recognizable.

Question 2 (1 point)

$$L_1 - L_2 = L_1 \cap \overline{L_2}.$$

Since L_2 may be RE but not recursive, $\overline{L_2}$ may not be recognizable, so their intersection with L_1 may not be recognizable. Other options either involve decidable complements or unions that preserve recognizability. Thus, only option B is not necessarily recognizable.

Question 3 (1 point)

Equivalence of two CFGs $L(G_1) = L(G_2)$ is undecidable, as it's impossible in general to determine if their languages are equal.

Equivalence for finite automata is decidable, as is membership for CFGs and acceptance by DFAs. Membership for CFGs and DFAs, and equivalence for DFAs, are all decidable problems. Only option D is undecidable by known theorems.

Question 4 (1 point)

If both a language and its complement are RE but not recursive, it contradicts the fact that if both are recognizable, then the language must be recursive.

Other cases — neither RE, one RE and one not, or both recursive — are possible.

Question 5 (1 point)

The set of Turing machines is countable, so every TM can be encoded by some integer — the mapping is onto. However, multiple encodings can represent the same TM, hence not one-one. Hence the mapping is onto but not one-one.

Question 6 (1 point)

Even if B is not recursively enumerable, A could still be recognizable. Even if B is not, A could still be RE. Thus, $A \leq_m B$ and B not RE does not necessarily imply A is not RE. The other statements are standard implications of many-to-one reducibility.

Question 7 (1 point)

In a deterministic TM, each configuration has at most one next step, so two nodes cannot share the same parent. Paths, cycles, and paths with cycles are allowed. Only trees with shared parents are invalid.

Question 8 (1 point)

Option A is true: Every non-deterministic TM can be simulated by a deterministic TM using standard enumeration techniques.

Option B is false: RE (Turing recognizable) languages are not closed under complementation. If both a language and its complement are RE, then the language is recursive, but this is not always the case.

Option C is true: Decidable languages are closed under intersection because we can decide membership in both languages.

Option D is true: RE languages are closed under union because we can simulate both recognizers in parallel and accept if either accepts.